

Generation Interconnection System Impact Study Report

For Queue Project Q11-0064

Parkin 138 kV

May 30, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0064. The Interconnection Customer proposes a 288.7 MW wind facility interconnecting at the Parkin 138 kV substation in the Virginia area. The request was received on August 16, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 52 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$34,185,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Parkin – Candler's Mtn 138kV
Contingency: Leesville – Reusens 345kV line
Loading: 169.8% (Rating: 177 MVA)
DFAX: 43.1%
Pre-project loading: 99.5%
2. Facility: South Lynchburg – Candler's Mtn 138kV
Contingency: Leesville – Reusens 345kV
Loading: 144.7% (Rating: 193 MVA)
DFAX: 43.1%
Pre-project loading: 80.3%
3. Facility: S Lynchburg – Brookville 138kV L
Contingency: Reusens – Leesville 345kV
Loading: 130.1% (Rating: 207 MVA)
DFAX: 43.1%
Pre-project loading: 70.1%
4. Facility: Boonsboro – Nacoochee 138kV
Contingency: Bear – Clover 345kV
Loading: 129.1% (Rating: 255 MVA)
DFAX: 22.5%
Pre-project loading: 103.6%
5. Facility: Opossum Crk Substation – Parkin 138kV
Contingency: Parkin – Candler's Mountain 138kV
Loading: 128.4% (Rating: 209 MVA)
DFAX: 100.0%
Pre-project loading: 9.8%
6. Facility: South Texa – Nacoochee 138kV
Contingency: Bear – Clover 345kV
Loading: 120.3% (Rating: 276 MVA)

DFAX: 22.5%

Pre-project loading: 96.8%

7. Facility: Central Tie – Opossum Crk 138kV line

Contingency: Parkin – Candler's Mtn 138kV

Loading: 119.8% (Rating: 216 MVA)

DFAX: 100.0%

Pre-project loading: 14.1%

8. Facility: S Creek – Wythe 138kV

Contingency: Parkin – Candler's Mountain 138kV

Loading: 113.6% (Rating: 195 MVA)

DFAX: 46.1%

Pre-project loading: 45.5%

9. Facility: Wythe – Central Tie Substation 138kV

Contingency: Parkin Substation – Candler's Mtn 138kV

Loading: 112.1% (Rating: 222 MVA)

DFAX: 100.0%

Pre-project loading: 18.2%

10. Facility: S Texa – Franklin-Elk Grove 138kV

Contingency: South Texa – Desert Outpost 138kV

Loading: 108.7% (Rating: 197 MVA)

DFAX: 15.6%

Pre-project loading: 85.9%

11. Facility: Bear Island – Dunn Siding 345kV

Contingency: Birchwood – Leavitt 345kV

Loading: 104.8% (Rating: 1,303 MVA)

DFAX: 14.1%

Pre-project loading: 101.6%

12. Facility: Amargo – Brookville 138kV

Contingency: Reusens – Leesville 345kV

Loading: 102.3% (Rating: 231 MVA)

DFAX: 43.1%

Pre-project loading: 48.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Parkin – Candler's Mtn 138kV	Rebuild line with higher-capacity conductor	\$1,396,000	100% (\$1,396,000)
South Lynchburg – Candler's Mtn 138kV	Rebuild line with higher-capacity conductor	\$4,532,000	100% (\$4,532,000)
Brookville – S Lynchburg 138kV	Rebuild line with higher-capacity conductor	\$3,240,000	100% (\$3,240,000)
Boonsboro – Nacoochee 138kV L	Rebuild line with higher-capacity conductor	\$4,306,000	76% (\$3,279,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Parkin – Opossum Creek 138kV	Rebuild line with higher-capacity conductor	\$2,582,000	100% (\$2,582,000)
South Texa – Nacoochee 138kV	Rebuild line with higher-capacity conductor	\$4,579,000	76% (\$3,487,000)
Central Tie Substation – Opossum Creek 138kV	Rebuild line with higher-capacity conductor	\$1,560,000	100% (\$1,560,000)
S Creek – Wythe 138kV	Rebuild line with higher-capacity conductor	\$8,996,000	64% (\$5,734,000)
Central Tie – Wythe 138kV	Rebuild line with higher-capacity conductor	\$3,385,000	100% (\$3,385,000)
Franklin-Elk Grove – S Texa 138kV L	Reconductor line	\$6,686,000	40% (\$2,703,000)
Dunn Siding – Bear Isand Substation 345kV	Reconductor line	\$27,876,000	7% (\$1,994,000)
Brookville – Amargo 138kV L	Reconductor line	\$293,000	100% (\$293,000)
	Total allocated to project		\$34,185,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.